Name: ______________________

SOLVING LINEAR EQUATIONS,
(EQUATIONS WITH BRACKETS)

Date: ________________

LO: I can solve linear equations that involve expanding brackets.

Solving equations with brackets

When an equation contains **brackets**, expand them first, then collect like terms and solve as normal. Remember to **multiply every term inside** the bracket.

Strategy: Expand all brackets, collect x terms on one side and numbers on the other, then divide.

Quick examples

🟢	2(x + 3) = 10 2x + 6 = 10 x = 2	expand then solve
🟢	3(2x - 1) = 15 6x - 3 = 15 x = 3	multiply both terms
🟢	5(x + 2) = 3(x + 6) 5x + 10 = 3x + 18 x = 4	brackets both sides
🔴	2(x + 3) = 2x + 3	must multiply both terms: 2x + 6

Key idea

Expand all brackets first, then solve by collecting like terms.

a(x + b) = c ax + ab = c

Multiply every term inside by the number outside

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – single bracket

Solve 3(x + 4) = 21

Expand: 3x + 12 = 21

Subtract 12: 3x = 9

Divide by 3: x = 3

$\boxed{x = 3}$

Expand first, then use inverse operations to isolate x.

Example 2 – bracket with subtraction

Solve 4(2x - 3) = 20

Expand: 8x - 12 = 20

Add 12: 8x = 32

Divide by 8: x = 4

$\boxed{x = 4}$

Remember to multiply the negative term too: 4 × (-3) = -12.

Example 3 – brackets on both sides

Solve 5(x + 1) = 2(x + 7)

Expand both: 5x + 5 = 2x + 14

Subtract 2x: 3x + 5 = 14

Subtract 5: 3x = 9, x = 3

$\boxed{x = 3}$

Expand both sides, then collect x terms together.

Example 4 – negative outside bracket

Solve 10 - 2(x + 3) = 0

Expand: 10 - 2x - 6 = 0

Simplify: 4 - 2x = 0

2x = 4, x = 2

$\boxed{x = 2}$

The negative sign multiplies both terms: -2 × x = -2x, -2 × 3 = -6.

YOUR TURN – PRACTICE (deliberate practice)

1. Single bracket
Expand and solve.

a) $2(x + 5) = 16$

b) $3(x + 2) = 15$

c) $5(x - 1) = 20$

d) $4(x - 3) = 8$

2. Coefficient of x not 1
Expand and solve.

a) $2(3x + 1) = 14$

b) $3(2x - 5) = 9$

c) $5(4x + 2) = 50$

d) $4(3x - 1) = 32$

3. Brackets on both sides
Expand both sides and solve.

a) $3(x + 2) = 2(x + 5)$

b) $4(x + 1) = 2(x + 9)$

c) $5(x - 3) = 3(x + 1)$

d) $6(x - 1) = 4(x + 2)$

4. Negative outside bracket
Be careful with the signs.

a) $12 - 3(x - 1) = 0$

b) $20 - 2(x + 3) = 4$

c) $15 - 4(x - 2) = 3$

d) $8 - (2x + 1) = 3$

5. Mixed bracket equations
Solve each equation.

a) $2(x + 4) + 3(x - 1) = 20$

b) $4(2x + 1) - 3(x - 2) = 25$

c) $7(x - 2) = 3(x + 2) + 6$

d) $5(x + 3) - 2(x + 1) = 22$

e) $3(2x + 5) = 2(4x - 1) + 9$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) $2(x + 3) = 2x + 6$ ☐
Correct answer: __________
Reason: __________________

b) $3(x - 2) = 3x - 2$ ☐
Correct answer: __________
Reason: __________________

c) Expand all brackets before solving ☐
Correct answer: __________
Reason: __________________

d) $-2(x + 3) = -2x + 6$ ☐
Correct answer: __________
Reason: __________________

e) $-2(x + 3) = -2x - 6$ ☐
Correct answer: __________
Reason: __________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Solve $\mathbf{3(2x + 1) - 2(3x - 4) = 5(x - 1)}$. Show clearly how you expand each bracket.

Expression: ____________________________

Simplified: ____________________________

b) A rectangle has length **(3x + 2) cm** and width **(x + 1) cm**. Its perimeter is **46 cm**. Write an equation and solve to find x, then find the dimensions of the rectangle.

Expression: ____________________________

Simplified: ____________________________

TOP TIPS
Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK
Have I shown all my working clearly?
Did I check my answer using a different method?
Does my answer look reasonable?

WELL DONE!